Real-Time Deformable-Contact-Aware Model Predictive Control for Force-Modulated Manipulation

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***Abstract*—The force modulation of robotic manipulators has been extensively studied for several decades. However, it is not yet commonly used in safety-critical applications due to a lack of accurate interaction contact modeling and weak performance guarantees—a large proportion of them concerning the modulation of interaction forces. This study presents a high-level framework for simultaneous trajectory optimization and force control of the interaction between a manipulator and soft environments, which is prone to external disturbances. Sliding friction and normal contact force are taken into account. The dynamics of the soft contact model and the manipulator are simultaneously incorporated in a trajectory optimizer to generate desired motion and force profiles. A constrained optimization framework based on the alternative direction method of multipliers has been employed to efficiently generate real-time optimal control inputs and high-dimensional state trajectories in a model-predictive control fashion. The experimental validation of the model performance is conducted on a soft substrate with known material properties using a Cartesian space force control mode. Results show a comparison of ground truth and real-time model-based contact force and motion tracking for multiple Cartesian motions in the valid range of the friction model. It is shown that a contact-model-based motion planner can compensate for frictional forces and motion disturbances and improve the overall motion and force tracking accuracy. The proposed high-level planner has the potential to facilitate the automation of medical tasks involving the manipulation of compliant, delicate, and deformable tissues.**

***Index Terms*—Adaptive control, data-driven modeling, force control, manipulators, trajectory optimization.**

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I. Introduction

ROBOTICS applications in the medical domain have gained increasing attention over the past few decades [1], [2], where the planning and control of interaction forces between a robot and its environment are essential to various safety-critical tasks. For instance, interaction forces should be modulated accurately in compliant environments, such as surgical settings, microassembly, or biological tissue manipulation. Furthermore, force control based on identifiable physical models is crucial to identify instability modes (e.g., those caused by the bandwidth and system structure) and maintain reliable force interaction to guarantee safety. Thus, a model-based trajectory planning method with a high-fidelity contact model is essential for successful deployment with satisfactory motion and contact force tracking performance.

In the field of model-based planning and simulation, there have been extensive studies in handling rigid contact and deformable contact models. For instance, rigid contact can be modeled using numerical solutions to complementarity problems [3], [4], [5], [6] or hybrid models with impact dynamics [7], [8]. Unlike in rigid body contact, soft contact models are subject to challenges posed by nonlinear material properties and nonuniformity as well as intensive computation burden due to numerical computation for solutions. Considering soft contact mechanics during a physical interaction is important for safe adaptation and planning, especially in robotic surgical applications. Soft contact modeling is challenging as it depends on material properties, deformation, and rolling friction. Numerous contact models have been presented in the literature to model interactions involving elastic deformation [9], [10], [11], [12], [13]. These models have broad applications and are essential in many engineering areas such as machine design, robotics, multibody analysis, to name a few. For contact problems that involve elasticity, Hertz adhesive contact theory has been well established [14]. In this study, we focus on robotic tasks interacting with soft tissues, the contact behavior of which is determined by not only external and viscous forces, and contact geometry, but also material properties (see Fig. 1).

Many robotic tasks require motion planning in the presence of contact in constrained environments. For example, in the automation of medical applications, the environment is heavily constrained and safety is paramount [15]. In such scenarios,

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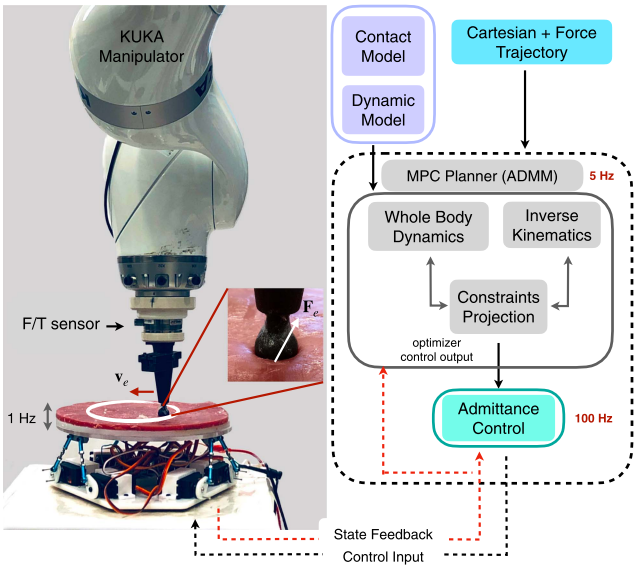


Fig. 1. KUKA manipulator experimentation setup, a manipulator performing a force-controlled motion task on a soft surface.

the robot motion and the reaction force imparted on the environment are coupled together. Similar examples are found in automated manufacturing, legged locomotion, and assistive robotics [16], [17], [18]. As a promising approach along this direction, trajectory optimization (TO) with either hard or soft complementarity constraints [3], [19], [20], [21], [22], [23], [24] or explicit contact models [25], [26], [27], [28], [29] has been extensively investigated in the robotics community. By incorporating the contact forces and joint states into the optimization, contact-dynamics-consistent motions can be planned for complex robot behaviors, such as dynamic locomotion or dexterous object manipulation. While the above works demonstrated impressive results on automatically discovering contact sequences, the optimized contact force was not used for control and only served as a way to explore the physical interaction with objects. In this article, we aim to accurately track a nominal force trajectory along with an end-effector path, which is commonly demanded but underexplored in safety-critical tasks.

Safety-critical tasks such as soft material manipulation and medical applications require extensive manipulation with accurate contact interactions. These include interaction with humans in proximity or with direct physical contact. In approaching the manipulation interaction problem, various methods have been explored, including model-based and model-free methods. Data-driven techniques have been studied to learn the interaction between robotic manipulators and the environment [30], [31], [32], [33]. Unlike rigid contacts, soft environments are prone to both spatial and temporal uncertainties, and it is challenging to learn the contact model and robot dynamics simultaneously through data [11], [34], [35]. In contrast, model-based approaches could capture the principal component of contact dynamics to a reasonable fidelity and be used along with an appropriate planning framework [36], [37], [38]. These studies present control frameworks and sensor modalities that incorporate contact models to make use to make informed control decisions, including physics-based contact models [36], mesh-based simulation models, and tactile models [37]. However, in these studies, simultaneous motion and force tracking is largely underexplored. Our method models the contact with normal and frictional components of the force as a smooth dynamical system to augment the robot dynamics. Then, we embed it into a contact-aware TO for high-fidelity planning.

Since contact-rich environments are prone to disturbances and difficult to model or predict, it is indispensable to solve the aforementioned optimization problem online with updated contact state information. However, efficiently solving highly constrained TO with full robot and contact dynamics remains an open problem. Instead of solving a large-scale optimization in a holistic way, we propose a distributed optimization framework inspired by works from the legged locomotion community [39], [40], [41], [42], [43], but employ a different way for decomposing the original problem given the soft manipulation task. More specifically, this framework efficiently alternates between three subproblems/subblocks: 1) an unconstrained subproblem that only incorporates coupled rigid body (i.e., the robot end-effector) and soft contact dynamics for accurate force tracking; 2) an inverse kinematics (IK) subproblem that generates a joint trajectory to track the desired end-effector trajectory; and 3) a constraint projection subproblem for handling inequality constraints including contact constraints. The proposed distributed framework enables an online execution in an model-predictive control (MPC) fashion. Furthermore, to maintain the contact and stability, an additional control layer will need to be deployed on top of the high-level trajectory planner. We add a low-level admittance force controller (FC) [44] to handle uncertainties arising from the model as well as from disturbances in the environment.

The main contributions of this article are listed as follows:

1) presentation of a dynamic interaction model based on soft contact mechanics for a predefined geometry with Hertz viscostatic theory;
2) incorporation of the interaction model into a constrained TO to generate the desired Cartesian path and force profile in an efficient distributed fashion;
3) experimental validation of the derived contact dynamical model and real-time implementation of the proposed TO algorithm with MPC.

A conference version of the work presented in this article was published in [45]. The work presented here extends the previous work in three respects. First, we extend the previous distributed framework into three subproblems instead of only two, i.e., an independent IK subproblem is extracted from the previous dynamics block to further split the dynamics (force tracking) and kinematics (end-effector tracking) planning. We benchmark different versions of solving the same constrained TO problem including the one proposed in our conference version [45]. Second, our proposed trajectory planner is executed online in an MPC fashion. We experimentally demonstrate the efficacy of this framework on simultaneous motion and force tracking tasks on a static and dynamic disturbance-induced platform.

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Last but not least, we implement a low-level controller to aid the high-level trajectory planner in motion and force trajectory tracking.

II. Related Work

A. Contact Models

Elastic contact mechanics [14] have been extensively studied in various research fields where contact modeling is imperative for safety and performance requirements. Existing works in [10], [46], [47], [48] have used soft contact models for both modeling and control. These works include quasi-static assumptions, and studies of [49], [50] explore cases where high-velocity impacts on soft material are considered. In the impact cases, viscoelastic models have been widely investigated. For instance, studies in [10], [49] compared various viscoelastic models with experimental validations. A majority of these works show that the Hertzian-based Hunt–Crossley model is the one most suitable for viscoelastic cases. Furthermore, fundamentals of frictional sliding motion are established in [13], [51], where the main focus is on rigid body contacts but generalizable to soft contacts. More recent works in [12], [52] propose contact-area-based models.

B. Trajectory Optimization

TO is a powerful tool to generate reliable and intelligent robot motions. Various numerical optimization methods have been proposed to solve a TO [53], [54], [55]. Among them, differential dynamic programming (DDP) [56] and its first-order variant iterative linear–quadratic regulator (iLQR) [57] have aroused much attention in solving TO in the context of unconstrained problems, where only dynamics constraint is enforced. The Ricatti-like backward pass in DDP or iLQR effectively reduces the complexity by solving an approximated linear–quadratic regulator problem over the entire horizon, and the optimization is solved in an iterative fashion. In [58], DDP is used in a balancing task of a humanoid robot with high degrees of freedom (DoFs). A follow-up work [59], demonstrates an MPC implementation based on DDP. However, incorporating additional constraints in standard DDP algorithms is still an open problem. In [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], DDP-type variants are proposed to cope with equality or inequality constraints containing state or control variables. Meanwhile, an augmented Lagrangian (AL)-based method, alternating direction method of multipliers (ADMM) [71], is proposed to solve the constrained optimization problem in a distributed fashion. In [72], [73], the ADMM is used to solve large-scale convex problems with box and cone constraints. Although the convergence is only guaranteed for the convex problem, the ADMM has been proven effective for solving highly nonconvex problems such as collision avoidance [74], [75], mixed-integer programming [76], [77], linear complementarity problem [78], and recently employed for legged robot trajectory planning [40], [41]. Our approach leverages both the distributed nature of ADMM and the efficiency of DDP, which decomposes the original problem into unconstrained whole-body dynamics planning with a soft contact model, kinematics planning, and constraint projection.

C. TO With Soft Contact

Contact-aware TO for legged locomotion and dexterous object manipulation is often built upon the assumption of rigid contact dynamics [3], [19], [20], [21], [22], [23], [25], [26], [27], [28], [29]. In [25], [26], [79], soft contact models are integrated into the system dynamics to approximate the hard contact, and the contact models are still relatively simple. In [80], a soft contact model was considered in the optimization formulation for whole-body locomotion control. Although incorporating these soft contact models has demonstrated impressive results, it is challenging to directly use these approaches for soft manipulation tasks considered in this article. Most of them assume simple spring-damper-type soft contact models, which still largely mismatch the contact surface deformation or elasticity [10], [36], [37], [38], [39] in reality. Therefore, advanced planning algorithms that accurately model complex contact dynamics are imperative to enable maneuvering over complex terrain or grasping irregular objects. To date, constrained TO incorporating a high-fidelity deformable contact model remains underexplored in the field.

D. Model-Predictive Control

The generation of trajectories for a given model and fixed horizon is computationally prohibitive in nature, making it difficult to deploy in real-time contact-rich applications, where model uncertainties and environmental disturbances are ubiquitous. MPC is a powerful strategy widely used to generate motion plans in real time and be adaptive to state changes due to environmental disturbances [81], [82]. Recent advances in fast automatic differentiation (AutoDiff) [83] and AutoDiff compatible rigid body models has enabled real-time optimal control. The study in [84] showed a hardware implementation of MPC with a DDP optimizer framework at an update rate of 1000 Hz on a seven-DoF robot for a vision based point-to-point trajectory planning.

E. Admittance Control

To cope with unmodeled modalities of the contact, we use a low-level FC based on admittance control [44], which has a fast control update rate compared to the high-level planner. It can compensate for the uncertainties that arise spatially over the surface (e.g., stiffness, slipperiness, and damping). Admittance control has been long studied and proven to work efficiently in compliant environments [85]. Furthermore, a low-level controller mitigates instabilities [86] arising from the contact caused by the control update rate, stiffness mismatch, and high gains.

III. Deformable Contact Modeling

A. Contact Modeling via Hertz's Theory

In this section, we model the interaction dynamics between an application tool mounted on a manipulator and a soft tissue in terms of contact geometry and mechanics. For simplicity, the tool-tip of this study is of a spherical shape (but not limited to). We assume that the tool end tip used is rigid and relatively stiffer compared to the contact surface. With these assumptions, we

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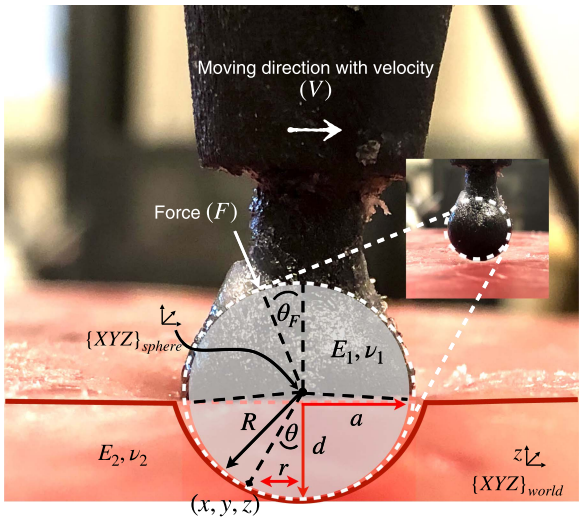


Fig. 2. Graphical illustration of the soft contact model between surface and the end-effector tool.

derive a contact dynamical model based on the contact friction theory and pressure distribution. According to Hertz's theory, the largest static deformation is observed at the midpoint of the circle (as shown in the deformation in Fig. 2) and can be expressed as

$$d = \left[\frac{9F^2}{16E^2R}\right]^{\frac{1}{3}} \tag{1}$$

where E is the reduced Young's modulus of the tool and surface, R is the radius of the tool end, and F is the force imparted on the surface by manipulator tool-tip. Combined Young's modulus of the tool-tip and the soft contact surface material can be lumped to one term as

$$\frac{1}{E} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2}$$

where E_1, E_2, and ν_1, ν_2 are Young's moduli and Poisson ratios of the end-effector and contact surface material, respectively. In our scenario, we assume the contact part as a rigid object, and thus, Young's modulus of the spherical tool-tip E_1 is comparatively high. Consequently, the lumped stiffness can be approximated as $E = E_2/(1-\nu_2^2)$.

The deformation and stress distributions on the contact surface are approximated by the universal Hooke's law and Hertz's theory. A detailed elaboration of normal, radical, and hoop (i.e., moving direction) stress distributions within the contact area in the cylindrical coordinate system are provided in Appendix A.

The deformation distribution is derived from the stress distribution equations as follows:

$$\sigma_n = \begin{cases} \frac{3\pi}{8a}\left[\frac{1-\nu^2}{E}\right] p_m(2a^2 - r^2), & (r \le a) \\ \frac{3}{4a}\left[\frac{1-\nu^2}{E}\right] p_m\left[(2a^2 - r^2)\sin^{-1}\left(\frac{a}{r}\right)\right. & \\ \quad \left. + a(r^2 - a^2)^{\frac{1}{2}}\right], & (r \ge a) \end{cases}$$

where $p_m = F/(\pi a^2)$ is the average stress applied in the contact part by the tool-tip and $a = \sqrt{Rd}$ is the radius of contact area (see Fig. 2). A force vector F at an angle θ_F to the normal is applied to the application tool and moves in a curved path of a radius R with a uniform velocity $\mathbf{v}_e$ in the frame {sphere}. This represents a scenario of a tool interaction with a soft surface to accomplish a motion task. For the sake of simplicity, our model focuses on sliding friction (primary mode) and ignores other frictional sources such as adhesion and rolling induced by deformation. This assumption is valid for modeling purposes where adhesion and rolling are secondary and specific to the tool material and the application.

To derive frictional forces, we use principal stress on the contact. If σ_θ represents the principal stress within the contact circle due to the symmetry of our contact scenario, we can represent the stress tensor of any contact point (r, θ, z) in cylindrical coordinates relative to frame {sphere} via the Cauchy stress theory [14]

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_r & 0 & \sigma_{rz} \\ 0 & \sigma_\theta & 0 \\ \sigma_{zr} & 0 & \sigma_z \end{bmatrix}. \tag{2}$$

Since the task is defined in the Cartesian frame, we convert parameters to Cartesian coordinates from cylindrical coordinates. The stress tensor in Cartesian coordinate is $\boldsymbol{\sigma}_c = T^T \boldsymbol{\sigma} T$, where the transformation matrix T is defined in Appendix A.

At an arbitrary point on contact surface $(x, y, z)_{\{\text{sphere}\}}$, the normal vector from this point to centroid of spherical cap is $\mathbf{n} = [s\theta \;\; 0 \;\; c\theta]^T$.[1] Then, the normal stress of the contact surface is $\sigma_n = \mathbf{n}^T \boldsymbol{\sigma}_c \mathbf{n}$ with

$$\sigma_n = \sigma_r c^2\theta s^2\theta + \sigma_\theta s^4\theta + \sigma_z c^2\theta + 2\sigma_z s\theta c^2\theta.$$

Given this stress expression, the overall friction force of the contact surface is represented as

$$df = \mu\sigma_n dS = \mu\sigma_n \times 2\pi r \frac{dr}{c\theta} \tag{3}$$

$$f = \int df c\theta = 2\pi\mu \int_0^a \sigma_n r dr \tag{4}$$

where df, dr, and dS are the differential elements of the friction, r, and contact area, respectively. In the surface normal direction, it is assumed that the surface is in contact with the end point of the tool. As a result, (1) always holds. To derive the dynamic model in the normal direction of contact, the derivative form of (1) is taken

$$\dot{z} = -\dot{d} = -\left[\frac{1}{6E^2RF_z}\right]^{\frac{1}{3}} \dot{F}_z \tag{5}$$

where z represents the position along the surface normal direction of the contact point, and the force along the normal direction is defined as $F_z = F\cos\theta_F$. In the moving direction, (4) and μF_z give the frictional force caused by the normal force F_z, which is

$$\mathbf{F}_f = f = \mu F_z \left[1 + (2\nu - 1)\frac{3a^2}{10R^2}\right]\mathbf{n}_v + k_d \mathbf{v}_e \tag{6}$$

where k_d is a damping coefficient in the moving direction. By substituting $a = \sqrt{Rd}$ and (6), we have the derivative form of

[1] we denote $\cos\theta = c\theta$ and $\sin\theta = s\theta$.

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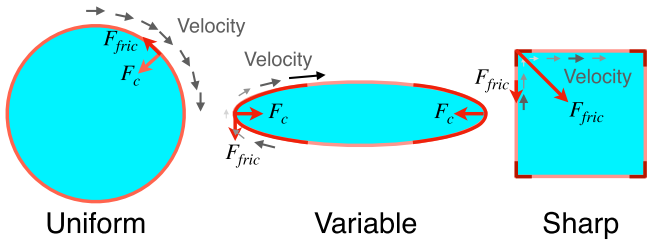


Fig. 3. Different Cartesian path geometries. At sharp corners of a Cartesian path, the robot needs to vary the velocity to compensate for centripetal and frictional force components caused through the contact.

(6). $\mathbf{n}_v$ is the unit vector of the velocity and $\mathbf{v}_e$ is the end-effector velocity at the contact point

$$\dot{\mathbf{F}}_f = \dot{f} = \mu \dot{F}_z + \frac{3\mu(2\nu-1)}{10R}\left(\dot{F}_z d - F_z \dot{z}\right)\mathbf{n}_v + k_d \dot{\mathbf{v}}_e. \quad (7)$$

The overall model, in simplistic terms, $\dot{\mathbf{F}}_e = \mathcal{G}(F_z, \mathbf{v}_e,$ contact parameters) with the frictional and normal force components of the contact model can be written in a compact form as

$$\dot{\mathbf{F}}_e = \left(\left(6E^2 R F_z\right)^{\frac{1}{3}} \dot{d}\right)\mathbf{n}_z + \left(\mu \dot{F}_z + \frac{3\mu(2\nu-1)}{10R}\left(\dot{F}_z d + F_z \dot{d}\right)\right)\mathbf{n}_v + k_d \dot{\mathbf{v}}_e \quad (8)$$

where $\mathbf{n}_v$ is a unit vector and d is the deformation at the central point of contact circle and is calculated from (1), and $\dot{d} = \dot{x}_z$. F_z is the vertical force (the surface normal direction) applied on the surface by the manipulator and $\mathbf{v}_e = ||\mathbf{J}\dot{\mathbf{q}}||$ is the moving velocity of the tool contact point.

B. Manipulator Dynamics

The manipulator model dynamics is expressed as follows:

$$\ddot{\mathbf{q}} = \mathbf{M}(\mathbf{q})^{-1}(\boldsymbol{\tau}_u - \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{G}(\mathbf{q}) - \mathbf{J}^T \mathbf{F}_e) \quad (9)$$

where $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \boldsymbol{\tau}_u \in \mathbb{R}^k$ are the manipulator joint position, velocity, acceleration, and torque vectors, respectively. We use $k = 7$ for our seven-DoF robotic arm. $\mathbf{M}(\mathbf{q})$ is the joint space mass matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ is the Coriolis term, $\mathbf{G}(\mathbf{q})$ is the gravity term, $\boldsymbol{\tau}_u$ is the torque applied at joints, $\mathbf{J}$ is the Jacobian with respect to the contact point, and $\mathbf{F}_e$ is the external Cartesian wrench at the end-effector.

C. Contact Constraint Modeling

The motion studied in this article is primarily in the sliding mode, which an equality constraint can describe. Since the sliding is embedded in the contact model, additional constraints for sliding are not required. However, a constraint is added to make sure that the robot is only sliding in the desired path where the path is curved. Fig. 3 shows the two components of the force acting on the tool, namely, frictional and centripetal forces along the path of curvature R_c, as shown in Fig. 4. The contact model provides the sliding friction, and the centripetal force constraint is added as a constraint

$$\frac{\mathbf{J}^{-T}\mathbf{M}(\mathbf{q})\mathbf{J}^{-1}\|\mathbf{J}\dot{\mathbf{q}}\|^2}{\kappa} \leq \mu \mathbf{N}^T \mathbf{F}_e \mathbf{N} \quad (10)$$

where $\mathbf{F}_e \in \mathbb{R}^3$ is the force vector at the end-effector and $\mathbf{J}$ is the Jacobian. κ is the curvature of the motion path (as shown in Fig. 4), $\mathbf{J}^{-T}\mathbf{M}(\mathbf{q})\mathbf{J}^{-1}$ is the effective mass at the contact point of the robot with a mass matrix of $\mathbf{M}(\mathbf{q})$ and $\mathbf{J}\dot{\mathbf{q}}$ is the moving velocity of the contact point. $\mathbf{N}$ is the surface average normal vector (as precept by the force–torque sensor). In this study, we use $\mathbf{N} = [0 \quad 0 \quad 1]^T$, which is only in the z-direction.[2] The constraint represented by (10) keeps the robot in contact when the Cartesian tracking path has a curvature, and the robot is operated in a lower impedance mode. It is an effect of the resulting centripetal force on the effective mass at the contact. For instance, if the velocity at a curve is high, it would slip in the orthogonal direction of the moving direction unless high positional gains are used to compensate for it.

IV. Problem Formulation

The optimization problem is to solve a control trajectory that would result in a desired Cartesian trajectory along a desired force profile. The overall problem is formulated in Formulation 1, where the state is represented as $\mathbf{x} = [\mathbf{q} \ \dot{\mathbf{q}} \ \mathbf{F}_e]^T$, including the manipulator joint position, velocity, and the end-effector force vectors. The control input is equivalent to the joint torque vector $\mathbf{u} = \boldsymbol{\tau}_u$. For simplicity, we use $\boldsymbol{\phi} = (\mathbf{x}[0, \ldots, N], \mathbf{u}[0, \ldots, N-1])$ to represent the sequence of state–control pairs.

The objective function is comprised of a force tracking cost, an end-effector pose tracking cost, and a regularization term on the applied torques. The normal force error with respect to the force reference F^d is expressed as $\delta\mathbf{F}[i] = (F_z[i] - F^d[i])$. Matrices $\mathbf{Q} \in \mathbb{R}^{n\times n}$ and $\mathbf{R} \in \mathbb{R}^{m\times m}$ are the state and control weighting matrices, respectively, $\mathcal{FK} \in \mathbb{R}^{4\times 4}$ is the forward kinematics function of the manipulator, and $W_p \in \mathbb{R}^{4\times 4}$ is the state weighting matrix for the pose tracking cost with respect to a desired end-effector trajectory $\mathbf{x}_e^d$.

After combining the contact dynamics model in (8) and the manipulator dynamics model in (9), the overall system dynamics can be discretized through the Euler method and written as function $\mathbf{f}(\cdot,\cdot)$ in (11b). The initial condition $\mathbf{x}_0$ is given as shown in (11c). The joint and torque limits are defined in (11d) and (11e) with lower and upper bound pairs $(\mathbf{q}_{\text{lower}}, \mathbf{q}_{\text{upper}})$ and $(\mathbf{u}_{\text{lower}}, \mathbf{u}_{\text{upper}})$, respectively. The sliding contact constraint in (11f), discussed in Section III-C, also needs to be incorporated into the optimization problem. Next, we describe how we solve Formulation 1 in a distributed manner, i.e., the proposed ADMM algorithm.

V. Constrained Trajectory Optimization With Contact Dynamics

Given the manipulator and the contact dynamics models, DDP is used to generate desired joint and Cartesian motion as well as

[2] $\mathbf{N}$ varies with the surface deformation, which is stochastic and can estimated through an external force–torque sensor attached to the end-effector

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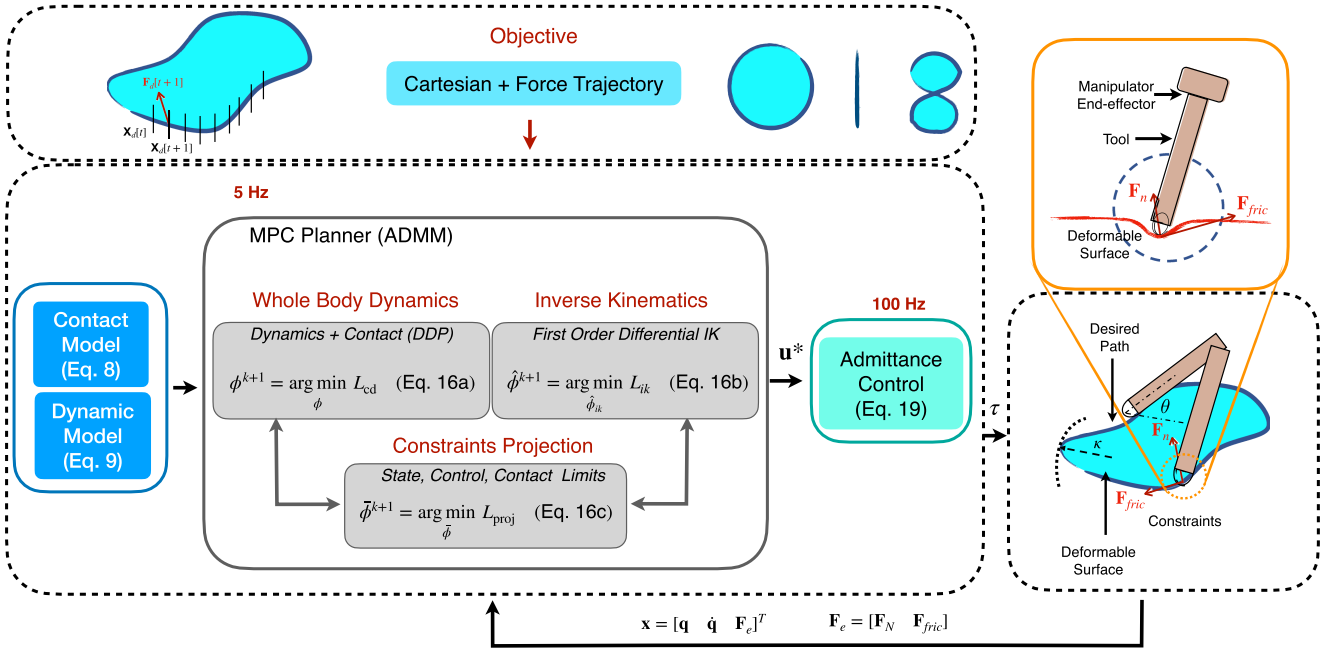


Fig. 4. Planning framework. The planner is composed of an ADMM-based solver executed in MPC fashion at 5 Hz. The ADMM solver contains three subblocks for optimization. Manipulator and contact dynamic models are used in the whole body dynamics block. Inverse kinematics block uses first-order differential kinematics to generate Cartesian trajectory. States are projected to satisfy the state and contact constraints in the projection block. The low-level FC, which is the admittance controller, is executed at 100 Hz. The desired Cartesian and force trajectory is tracked while obeying the constraints provided.

Formulation 1: Simultaneous Trajectory and Force Optimization.

$$\text{(Tracking task)} \quad \min_{\phi} \sum_{i=0}^{N} \overbrace{\delta\mathbf{F}[i]^T \mathbf{Q}_F\, \delta\mathbf{F}[i]}^{\text{force tracking}} + \mathbf{u}[i]^T \mathbf{R}\mathbf{u}[i] + \overbrace{W_p \left\| \mathcal{FK}(\mathbf{q}[i]) - \mathbf{x}_e^d[i] \right\|_2^2}^{\text{pose tracking}}$$

$$\text{(Decision variables)} \quad \phi[i] = [\overbrace{[\mathbf{q}[i] \;\; \dot{\mathbf{q}}[i] \;\; \mathbf{F}_e[i]]}^{\mathbf{x}[i]}]^T, \mathbf{u}[i]^T]^T \quad \forall i = 1, \ldots, N-1 \tag{11a}$$

$$\text{(Dynamics)} \quad \text{s.t.} \quad \mathbf{x}[i+1] = \mathbf{f}(\mathbf{x}[i], \mathbf{u}[i]) \tag{11b}$$

$$\text{(Initial condition)} \quad \mathbf{x}[0] = \mathbf{x}_0 \tag{11c}$$

$$\text{(Joint limits)} \quad \mathbf{q}_{\text{lower}} \leq \mathbf{q}[i] \leq \mathbf{q}_{\text{upper}} \tag{11d}$$

$$\text{(Torque limits)} \quad \mathbf{u}_{\text{lower}} \leq \mathbf{u}[i] \leq \mathbf{u}_{\text{upper}} \tag{11e}$$

$$\text{(Contact Constraint)} \quad \frac{\mathbf{J}^{-T}\mathbf{M}(\mathbf{q})\mathbf{J}^{-1}\|\mathbf{J}\dot{\mathbf{q}}\|^2}{\kappa} \leq \mu \mathbf{N}^T \mathbf{F}_e \mathbf{N} \tag{11f}$$

force profiles obeying the system dynamics defined by $\mathbf{f}(\cdot,\cdot)$. DDP is well received for effectively solving unconstrained TO [54]. It represents an indirect method that only optimizes control inputs, and the dynamics constraint is implicitly satisfied during the forward trajectory rollout. Given an initial guess of control inputs, an updated state trajectory is generated by forward propagating the differential equation of system dynamics. Then, a quadratic approximation is constructed for the cost function and dynamics around the current trajectory, so that a Riccati recursion can be used to derive the optimal feedback control law. By iteratively updating the state and control trajectories, the optimization will converge to an optimal solution.

One limitation of DDP stems from its difficulty in addressing constraints other than the dynamics constraint enforced during the forward pass rollout. Since our contact model enforces state, control, and frictional constraints, it is desired to incorporate these contact constraints along with the state and control constraints. Our previous work in [45] proposed an iterative and distributed method based on the ADMM to incorporate the contact dynamics and additional constraints. In this article, we introduce an IK subproblem and extend the entire ADMM framework to be a consensus variant to further improve the computational efficiency. Note that a sequential variant can also be established. Details are demonstrated in Appendix B, and we benchmark multiple variants in Section VII-A.

The ADMM algorithm decomposes a large-scale holistic optimization problem into subproblems and solves each subproblem iteratively. In each iteration, the primal and dual problems are solved sequentially. Under mild conditions, all the variable sets coming from primal and dual problems converge to the optimal solutions. More details about the ADMM algorithm are referred to [71]. To apply this algorithm for our soft-contact manipulation problem, we define various sets of copies and the corresponding consistency constraints. Therefore, the original

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Formulation 2: Distributed Constrained Optimization (Consensus).

$$
\begin{aligned}
&\text{(Tracking task)} && \min_{\boldsymbol{\phi},\hat{\boldsymbol{\phi}},\bar{\boldsymbol{\phi}}} \sum_{i=0}^{N} \delta\mathbf{F}[i]^T\mathbf{Q}_F\,\delta\mathbf{F}[i] + \mathbf{u}[i]^T\mathbf{R}\mathbf{u}[i] \\
&&& \quad + W_p\left\|\mathcal{FK}(\mathbf{q}[i]) - \mathbf{x}_e^d[i]\right\|_2^2 \\
&&& \quad + I_{\mathcal{J},\mathcal{U},\mathcal{F}}(\bar{\mathbf{q}}[i],\bar{\mathbf{u}}[i],\bar{\boldsymbol{\lambda}}[i]) \\
&\text{(Variables-DDP)} && \boldsymbol{\phi}[i] = [[\overbrace{\mathbf{q}[i]\ \dot{\mathbf{q}}[i]\ \mathbf{F}_e[i]}^{\mathbf{x}[i]}]^T, \mathbf{u}[i]^T]^T \\
&\text{(Variables-IK)} && \hat{\boldsymbol{\phi}}[i] = \hat{\mathbf{q}}[i]^T \\
&\text{(Variables-Proj)} && \bar{\boldsymbol{\phi}}[i] = [\bar{\mathbf{q}}[i]^T, \bar{\mathbf{u}}[i]^T, \bar{\boldsymbol{\lambda}}[i]^T]^T \\
&&& \forall i = 0, 1, \ldots, N-1 && \text{(12a)} \\
&\text{(Dynamics)} && \text{s.t.}\quad \mathbf{x}[i+1] = \mathbf{f}(\mathbf{x}[i], \mathbf{u}[i]) && \text{(12b)} \\
&\text{(Initial condition)} && \mathbf{x}[0] = \mathbf{x}_0 && \text{(12c)} \\
&\text{(Consistency constraints)} && \begin{cases} \hat{\mathbf{q}} = \bar{\mathbf{q}} \\ \mathbf{q} = \bar{\mathbf{q}} \\ \mathbf{u} = \bar{\mathbf{u}} \\ \boldsymbol{\lambda} = \bar{\boldsymbol{\lambda}} \end{cases} && \text{(12d)}
\end{aligned}
$$

Formulation 1 is transcribed into a distributed version as shown in Formulation 2. For simplicity, we define $\boldsymbol{\lambda} = (\dot{\mathbf{q}}^T, {\mathbf{F}_e}^T)^T$ and $\bar{\boldsymbol{\phi}} = (\bar{\mathbf{q}}[0,\ldots,N], \bar{\mathbf{u}}[0,\ldots,N-1], \bar{\boldsymbol{\lambda}}[0,\ldots,N])$[3] to express the concatenated copies of states and controls that are required to be projected. Meanwhile, the variable set $\hat{\boldsymbol{\phi}} = (\hat{\mathbf{q}}[0,\ldots,N])$ denotes a copy of joint position $\mathbf{x}_M$ and handles the end-effector tracking, i.e., the IK. The closed and convex sets $\mathcal{J}, \mathcal{U}$, and $\mathcal{F}$ stand for joint limit (11d), control limit (11e), and contact constraint (11f), respectively. By utilizing an indicator function that specifies a zero value if a targeting variable is within a feasible set and an infinitely large value if not, the above constraints are encoded as a cost term $I_{\mathcal{J},\mathcal{U},\mathcal{F}}$ inside the new cost function in Formulation 2.

Given the transcribed optimization problem, an AL in a scaled form (see [71, Sec. 3.1.1]) can be derived and written as follows:

$$
\begin{aligned}
\mathcal{L}(\phi,\hat{\phi},\bar{\phi},\mathbf{v}) = &\sum_{i=0}^{N}(\delta\mathbf{F}[i]^T\mathbf{Q}_F\delta\mathbf{F}[i] + \mathbf{u}[i]^T\mathbf{R}\mathbf{u}[i]) \\
&+ W_p\left\|\mathcal{FK}(\hat{\mathbf{q}}[i]) - \mathbf{x}_e^d[i]\right\|_2 \\
&+ I_{\mathcal{J},\mathcal{U},\mathcal{F}}(\bar{\mathbf{q}}[i],\bar{\mathbf{u}}[i],\bar{\boldsymbol{\lambda}}[i]) \\
&+ \frac{\rho_j}{2}\|\hat{\mathbf{q}} - \bar{\mathbf{q}} + \mathbf{v}_{\text{ik}}\|_2^2 + \frac{\rho_j}{2}\|\mathbf{q} - \bar{\mathbf{q}} + \mathbf{v}_j\|_2^2 \\
&+ \frac{\rho_u}{2}\|\mathbf{u} - \bar{\mathbf{u}} + \mathbf{v}_u\|_2^2 + \frac{\rho_f}{2}\|\boldsymbol{\lambda} - \bar{\boldsymbol{\lambda}} + \mathbf{v}_f\|_2^2
\end{aligned}
$$

[3]The decision variables $\mathbf{q}$ and $\boldsymbol{\lambda}$ are subsets of the full state $\mathbf{x}$.

where ϕ, $\hat{\phi}$, and $\bar{\phi}$ are primal variables. $\mathbf{v}_{\text{ik}}$, $\mathbf{v}_j$, $\mathbf{v}_u$, and $\mathbf{v}_f$ are dual variables related to each consistency constraints defined in (12d). ρ_j, ρ_u, and ρ_f are step-size parameters corresponding to each constraint. Note that since the first two consistency constraints possess the same projection goal $\bar{\mathbf{x}}_M$, parameter ρ_j is adopted for both the constraints.

Then, based on the ADMM, the original problem is divided into three subproblems, which are also known as subblocks. Each subblock only requires part of the aforementioned AL as the local cost function:

Whole-body dynamics subblock:

$$
\begin{aligned}
\mathcal{L}_{\text{wbd}}(\phi,\bar{\phi},\mathbf{v}) = &\sum_{i=0}^{N}(\delta\mathbf{F}[i]^T\mathbf{Q}_F\,\delta\mathbf{F}[i] + \mathbf{u}[i]^T\mathbf{R}\mathbf{u}[i]) \\
&+ \frac{\rho_j}{2}\|\mathbf{q} - \bar{\mathbf{q}} + \mathbf{v}_j\|_2^2 + \frac{\rho_u}{2}\|\mathbf{u} - \bar{\mathbf{u}} + \mathbf{v}_u\|_2^2 \\
&+ \frac{\rho_f}{2}\|\boldsymbol{\lambda} - \bar{\boldsymbol{\lambda}} + \mathbf{v}_f\|_2^2 \qquad (13)
\end{aligned}
$$

Inverse Kinematics subblock:

$$
\begin{aligned}
\mathcal{L}_{\text{ik}}(\hat{\phi},\bar{\phi},\mathbf{v}) = &\sum_{i=0}^{N} W_p\left\|\mathcal{FK}(\hat{\mathbf{q}}[i]) - \mathbf{x}_e^d[i]\right\|_2 \\
&+ \frac{\rho_j}{2}\|\hat{\mathbf{q}} - \bar{\mathbf{q}} + \mathbf{v}_{\text{ik}}\|_2^2 \qquad (14)
\end{aligned}
$$

Projection subblock:

$$
\begin{aligned}
\mathcal{L}_{\text{proj}}(\phi,\hat{\phi},\bar{\phi},\mathbf{v}) = &\sum_{i=0}^{N} I_{\mathcal{J},\mathcal{U},\mathcal{F}}(\bar{\mathbf{q}}[i],\bar{\mathbf{u}}[i],\bar{\boldsymbol{\lambda}}[i]) \\
&+ \frac{\rho_j}{2}\|\hat{\mathbf{q}} - \bar{\mathbf{q}} + \mathbf{v}_{\text{ik}}\|_2^2 + \frac{\rho_j}{2}\|\mathbf{q} - \bar{\mathbf{q}} + \mathbf{v}_j\|_2^2 \\
&+ \frac{\rho_u}{2}\|\mathbf{u} - \bar{\mathbf{u}} + \mathbf{v}_u\|_2^2 + \frac{\rho_f}{2}\|\boldsymbol{\lambda} - \bar{\boldsymbol{\lambda}} + \mathbf{v}_f\|_2^2. \qquad (15)
\end{aligned}
$$

Then, for each ADMM iteration k, the updating sequence in a scaled form is

$$
\begin{aligned}
&\phi^{k+1} = \arg\min_{\phi} \mathcal{L}_{\text{wbd}}(\phi, \bar{\phi}^k, \mathbf{v}^k) && \text{(16a)} \\
&\qquad \text{s.t. (12b)} \\
&\hat{\phi}^{k+1} = \arg\min_{\phi_{\text{ik}}} \mathcal{L}_{\text{ik}}(\hat{\phi}, \bar{\phi}^k, \mathbf{v}^k) && \text{(16b)} \\
&\bar{\phi}^{k+1} = \arg\min_{\bar{\phi}} \mathcal{L}_{\text{proj}}(\phi^{k+1}, \hat{\phi}^{k+1}, \bar{\phi}, \mathbf{v}^k) && \text{(16c)} \\
&\mathbf{v}_{\text{ik}}^{k+1} = \mathbf{v}_{\text{ik}}^k + \hat{\mathbf{q}}^{k+1} - \bar{\mathbf{q}}^{k+1} && \text{(16d)} \\
&\mathbf{v}_j^{k+1} = \mathbf{v}_j^k + \mathbf{q}^{k+1} - \bar{\mathbf{q}}^{k+1} && \text{(16e)} \\
&\mathbf{v}_u^{k+1} = \mathbf{v}_u^k + \mathbf{u}^{k+1} - \bar{\mathbf{u}}^{k+1} && \text{(16f)} \\
&\mathbf{v}_f^{k+1} = \mathbf{v}_f^k + \boldsymbol{\lambda}^{k+1} - \bar{\boldsymbol{\lambda}}^{k+1}. && \text{(16g)}
\end{aligned}
$$

It is worth noting that both (16a) and (16b) only rely on the solutions from the previous ADMM iteration k, which means that the above two steps can be performed in parallel compared with the sequential update shown in Appendix B.

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Algorithm 1: ADMM Trajectory Optimization.

$\boldsymbol{\phi} \leftarrow \boldsymbol{\phi}^0, \hat{\boldsymbol{\phi}} \leftarrow \hat{\boldsymbol{\phi}}^0, \bar{\boldsymbol{\phi}} \leftarrow \bar{\boldsymbol{\phi}}^0$
$\mathbf{v}_j \leftarrow \mathbf{v}_j^0, \mathbf{v}_{\text{ik}} \leftarrow \mathbf{v}_{\text{ik}}^0, \mathbf{v}_u \leftarrow \mathbf{v}_u^0, \mathbf{v}_f \leftarrow \mathbf{v}_f^0$
repeat
  $\boldsymbol{\phi} \leftarrow \text{DDP}\ (\boldsymbol{\phi}, \bar{\mathbf{q}} - \mathbf{v}_j, \bar{\mathbf{u}} - \mathbf{v}_u, \bar{\boldsymbol{\lambda}} - \mathbf{v}_f)$ (16a)
  $\hat{\boldsymbol{\phi}} \leftarrow \text{IK}\ (\bar{\mathbf{q}} - \mathbf{v}_{\text{ik}})$ (16b)
  $\bar{\boldsymbol{\phi}} \leftarrow \text{Projection}\ (\frac{1}{2}(\hat{\mathbf{q}} + \mathbf{v}_{\text{ik}} + \mathbf{q} + \mathbf{v}_j), \mathbf{u} + \mathbf{v}_u, \boldsymbol{\lambda} + \mathbf{v}_f)$ (16c)
  $\mathbf{v}_{\text{ik}} \leftarrow \mathbf{v}_{\text{ik}} + \hat{\mathbf{q}} - \bar{\mathbf{q}}$
  $\mathbf{v}_j \leftarrow \mathbf{v}_j + \mathbf{q} - \bar{\mathbf{q}}$
  $\mathbf{v}_u \leftarrow \mathbf{v}_u + \mathbf{u} - \bar{\mathbf{u}}$
  $\mathbf{v}_f \leftarrow \mathbf{v}_f + \boldsymbol{\lambda} - \bar{\boldsymbol{\lambda}}$
until stopping criterion is satisfied
return $\boldsymbol{\phi}$

To efficiently solve the constrained optimization problem in (16a), DDP is deployed and the state trajectory is always dynamically feasible by performing the forward pass. For (16c), this minimization problem reduces to a projection operator on convex sets $\mathcal{J}, \mathcal{U}$, and $\mathcal{F}$

$$\begin{aligned}\bar{\boldsymbol{\phi}}^{k+1} = \arg\min_{\bar{\boldsymbol{\phi}} \in C} & \frac{\rho_j}{2}\|\hat{\mathbf{q}}^{k+1} - \bar{\mathbf{q}} + \mathbf{v}_{\text{ik}}^k\|_2^2 \\ & + \frac{\rho_j}{2}\|\mathbf{q}^{k+1} - \bar{\mathbf{q}} + \mathbf{v}_j^k\|_2^2 \\ & + \frac{\rho_u}{2}\|\mathbf{u}^{k+1} - \bar{\mathbf{u}} + \mathbf{v}_u^k\|_2^2 + \frac{\rho_f}{2}\|\boldsymbol{\lambda}^{k+1} - \bar{\boldsymbol{\lambda}} + \mathbf{v}_f^k\|_2^2 \\ C = \{(\bar{\mathbf{q}}, \bar{\mathbf{u}}, \bar{\boldsymbol{\lambda}}) | & \bar{\mathbf{q}} \in \mathcal{J}, \bar{\mathbf{u}} \in \mathcal{U}, \bar{\boldsymbol{\lambda}} \in \mathcal{F}\}.\end{aligned}$$

Then, a saturation function can be used to efficiently project the infeasible values onto the boundaries induced by different constraints

$$\bar{\boldsymbol{\phi}}^{k+1} = \Pi_{\mathcal{J},\mathcal{U},\mathcal{F}}\left[\frac{1}{2}(\hat{\mathbf{q}}^{k+1} + \mathbf{v}_{\text{ik}}^k + \mathbf{q}^{k+1} + \mathbf{v}_j^k), \mathbf{u}^{k+1} + \mathbf{v}_u^k, \boldsymbol{\lambda}^{k+1} + \mathbf{v}_f^k\right]. \quad (17)$$

The whole process of our ADMM algorithm is shown in Algorithm 1. The selection of $\bar{\boldsymbol{\phi}}$ and dual variable $\mathbf{v}$ is arbitrary, and we initialized them to be zero. The initial trajectory of $\boldsymbol{\phi}$ is generated by running forward dynamics with an initial guess of controls. The functions $DDP(\cdot)$ and $IK(\cdot)$ correspond to the implementations of our DDP solver and IK solver. In each ADMM iteration, the controls from last ADMM iteration will be sent to the current DDP solver as a warm-start, which makes the DDP solver converge faster within around ten iterations in each ADMM iteration after the initial one. Then, the trajectories are solved iteratively until a stopping criterion with regard to primal residuals (see [71, Sec. 3.3]) is satisfied (residuals of magnitude 10^{-2}).

VI. Model-Predictive Control

We implement the ADMM planner in an MPC fashion in the real-time deployment. At each MPC cycle, the optimization in ((11)) is solved with a horizon $H = N_{\text{steps}}\delta t$, and its solution is used as the warm-start for the next MPC cycle, i.e., $(\mathbf{x}_0, \mathbf{u}_0) \leftarrow (\mathbf{x}[i], \mathbf{u}[i])$, where $\mathbf{x}_0$ is replaced by some initial state $\mathbf{x}[i]$. The control input to the robot is computed as follows:

$$\begin{aligned}\mathbf{u}[i]^* &= \mathbf{u}[i] + PD(\mathbf{x}[i], \mathbf{x}^{\text{curr}}) \\ &= \overbrace{\mathbf{u}[i]}^{ff} + \overbrace{\mathbf{K}(\mathbf{q}^{\text{curr}} - \mathbf{q}[i])}^{fb} + \overbrace{\delta\mathbf{u}_{FC}(\mathbf{F}_e[i], \mathbf{F}_e^{\text{curr}})}^{fc} \quad (18)\end{aligned}$$

where $\mathbf{x}^{\text{curr}} = [\mathbf{q}^{\text{curr}}\ \dot{\mathbf{q}}^{\text{curr}}\ \mathbf{F}_e^{\text{curr}}]^T$ is the filtered (low-pass) current state and $\mathbf{K} \in \mathbb{R}^{n \times n}$ is a gain matrix. $\mathbf{u}[i]$ is composed of three terms: the feedforward (ff), feedback (fb), and the admittance FC (fc).

Remark 1: The low-level controller in the KUKA Sunrise software [87] uses user-defined joint space impedance control internally. We set a safe () robot impedance to execute our commands as well as to allow the manipulator to behave safely in the compliant environment.

The high-level MPC loop is implemented at 5 Hz with a horizon of 1 s and a time step of $dt = 0.02$ s. In each MPC iteration, we solve for a trajectory by the ADMM planner, as described in Section V, and we use C++11 in all of our code implementations.[4] To maintain the contact force accuracy and to avoid instabilities resulting from low frequency control update rate [44], [88], we use a low-level FC, which runs at 100 Hz, where the update rate is appropriate for the compliant (low-stiffness) environment.

In one of the ADMM planner blocks, we use DDP to solve for the robot and contact dynamics, which consumes most of the computational power. To make the real-time implementation feasible, we use AutoDiff for derivatives provided by CppADCodeGen [89], and RobCoGen [83] is used to derive the analytical rigid body dynamic model. On a Linux machine (Intel i7) with 3.4-GHz clock speed, average computation time for a horizon of 1 s took 150 ms. Without AutoDiff and analytical models, it takes 1300 ms. With AutoDiff, it is possible to run MPC at an approximate rate of 5 Hz.

Each iteration of the DDP takes on average 10 ms, while we limit the number of DDP iterations to 10 per ADMM cycle to ensure the solver returns an optimal trajectory on time. We run a maximum of five ADMM iterations in each trajectory computation cycle. While the constraint residuals are not guaranteed to reach the same threshold every trajectory iteration, our experimental results demonstrated that five ADMM iterations were sufficient to reduce residuals to the order of 10^{-2}. In an ideal MPC setup, the first control input from the output trajectory is applied to the robot, and the current state is used as the initial state for the MPC computation at the next iteration. However, there could be a significant computation burden for the trajectory planner in a practical setting. To take into account the computation and communication delay in the hardware system, we use an asynchronous MPC similar to [90], as shown in Fig. 5 The current state ($\mathbf{x}[t]$) is applied to the planner. It takes T time steps to compute the trajectory. During that period, the previous trajectory is executed. After T time steps, trajectory $\mathbf{U}_{t,t+H} = \{\mathbf{u}_t, \ldots, \mathbf{u}_{t+H}\}$ is returned by the planner. In the

[4]The code implementation can be found at https://github.com/lasithagt/admm

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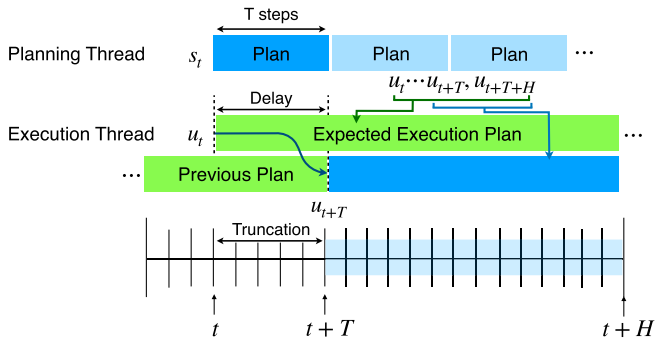


Fig. 5. MPC implementation. H, T, and $\mathbf{s}_t$ are the number of time steps for the horizon, time steps for the compute delay, and current state from the system, respectively. Computation takes T steps. During the time, execution thread keeps executing the previous plan. Once planning thread is finished, the current execution plan is updated; the time steps corresponding to delay is truncated.

actual execution of the trajectory, we truncate the control input sequence up to T time steps, as shown in Fig. 5. The planner and the execution threads are run in parallel in different threads using the threading library in C++11.

Although MPC is beneficial in handling model uncertainties and environmental disturbances, this is not sufficient for tasks that require force modulation. This is due to the instabilities that could arise from contact and determined by the control update rate and surface material properties [44], [88]. We use a FC that updates control input at a rate of 100 Hz to avoid contact-induced instabilities. Admittance control is proven [85] to be better suited for compliant environments where impedance control is best suited for stiff environments.

Force control is important as the surface parameters are not uniform and prone to unmodeled dynamics (e.g., damping, restitution, and slipping). To compensate for it, we use admittance control as the FC in the low-level control. To mitigate the instabilities that would arise from position-based admittance control [85], we use torque as the control input as opposed to position control. The stiffness of the contact surface material was estimated, as described in [44].

$$\boldsymbol{\delta}\mathbf{u}_{\mathrm{FC}}[t] = \mathbf{C}(x, y, z)\, \mathbf{J}^T(\mathbf{F}_e[t] - \mathbf{F}_e^{\mathrm{curr}}[t]) \tag{19}$$

where $\mathbf{J}(\mathbf{q})$ is the kinematic Jacobian matrix, $\boldsymbol{\delta}\mathbf{u}_{\mathrm{FC}}[t]$ is joint space torque, and $\mathbf{C}(x, y, z)$ is the compliance matrix that can vary spatially (in the space (x, y, z)).[5]

VII. Experiments

To validate the applicability of theoretical attributes of our work in a practical setting, we demonstrate it via physical experiments on a custom-designed platform, shown in Fig. 6 Moreover, we compare our framework with other state-of-the-art methods, which are used widely in robotics motion planning in simulation (in MATLAB©). Then, we show the contact parameter identification methods and compare results with the disturbance-induced tracking task.

[5]$\mathbf{C}$ is set to be a constant in this study and identified experimentally.

A. TO Algorithm Comparisons

The primary motivation for a distributed motion planning scheme such as ADMM is to use different optimization methods that specialize for each subproblem and incorporate constraints into it. For example, it is efficient to use first-order differential methods to generate IK solutions and solve for the dynamic model separately to find a consensus between them. These were benchmarked in MATLAB© with a 3.4-GHz i7-core processor. In addition, we benchmark our method against other standard methods.

1) *Our method [ADMM with three-block architecture (consensus)]:* The ADMM architecture with three blocks, i.e., nonlinear dynamics, IK, and projection blocks, is solved and consensus is found in the ADMM update. More details can be found in Appendix B.
2) *Sequential quadratic programming (SQP):* Direct collocation with trapezoidal transcription was used with the dynamical system with contact dynamics.
3) *iLQR—vanilla DDP:* iLQR [57] was used with a combined cost of the desired Cartesian trajectory ($SE(3)$), force trajectory, state, control, and contact constraints.
4) *ADMM with two-block architecture (sequential):* The ADMM scheme with two blocks, as implemented in [45], was used. In the nonlinear dynamics block, the desired state and control cost is used in the DDP solver. The projection block projects state, control, and contact constraints.
5) *ADMM with three-block architecture (sequential):* The ADMM architecture with three blocks, i.e., nonlinear dynamics, IK, and projection blocks, is solved sequentially in the ADMM update. More details can be found in Appendix B.

Fig. 7 shows the comparison of cost reduction and the contact residual reduction for the methods listed above. For the ADMM variants, the x-axis represents the number of dynamic solver (DDP) iterations instead of ADMM iterations for a fair comparison with other methods (e.g., SQP and iLQR).

B. Computation Time Benchmarking

Table I summarizes the computation time for each algorithm. The algorithms are implemented offline in MATLAB for the whole trajectory and derivatives taken with finite differences. It is observed that three-block consensus takes the shortest time, while two-block sequential takes the longest time. While the cost and residual convergence of SQP are comparable with three-block consensus, it takes about twice the time as three-block consensus. Vanilla DDP, where the cost is lumped altogether, has a comparable computation time to three-block consensus, but the residual cost does not decay over time as fast as other methods. Overall, the three-block consensus method renders the best computation time with a fast residual error decay rate.

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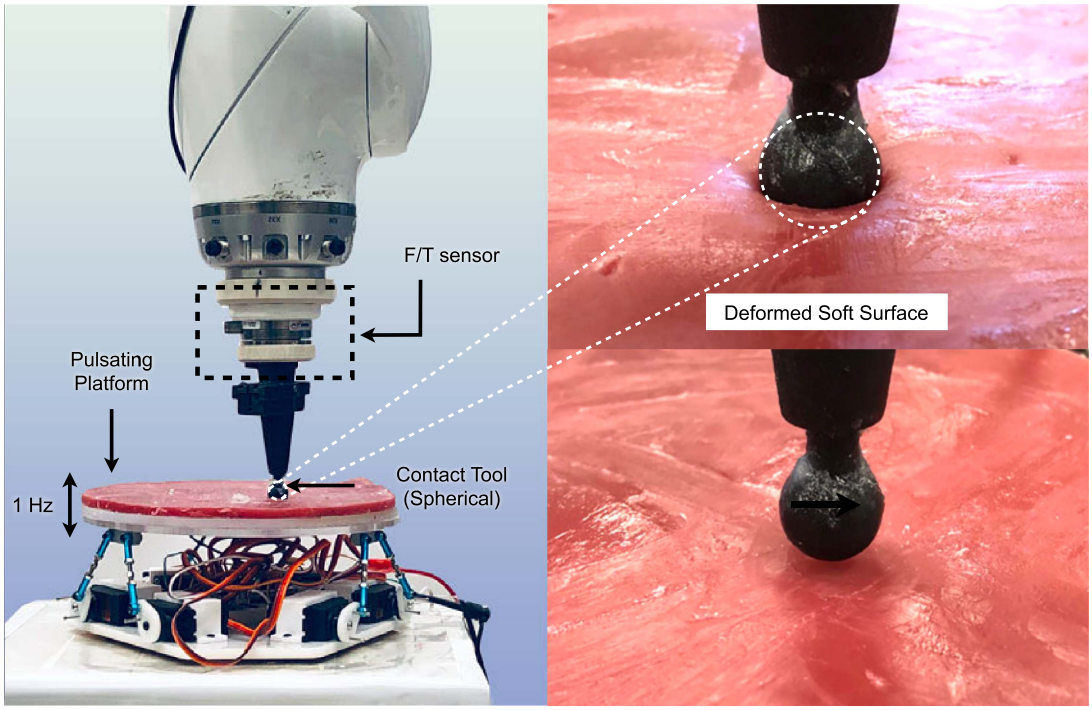


Fig. 6. Experimental setup. KUKA manipulator is equipped with spherical tool and a ATI force/torque sensor distally to the contact point. The pulsation platform moves up and down at a fixed rate of 1 Hz. Surface is covered with a thick soft substrate. The tool of the end-effector is comprised of a spherical indenter, which is in contact with the deformable surface.

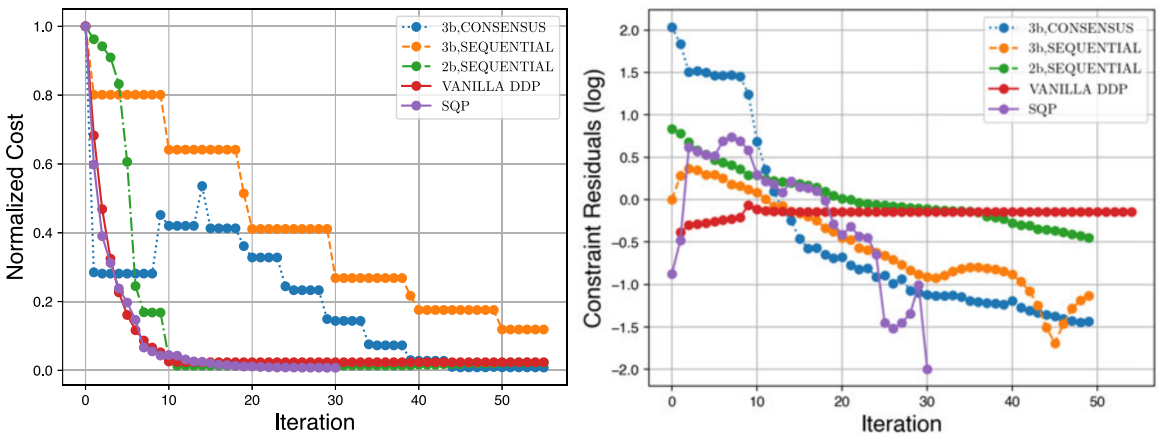


Fig. 7. Comparison of optimization methods. Three-block consensus, three-block sequential, two-block sequential, vanilla-DDP, SQP. *left:* Cost versus iterations. The cost is normalized such that the initial value is scaled to be unity. Vanilla DDP and SQP convergence are comparable and three-block consensus is the fastest. *right:* Constraint residuals versus Iterations. Three-block consensus residual convergence is faster although the initial constraint violation is high.

C. Cost Reduction

One advantage of using ADMM with a three-block architecture is to track a reference Cartesian trajectory ($SE(3)$) without adding an extra cost term in the dynamics block, which could impede the fast convergence of DDP. In this approach, the Cartesian trajectory can be solved efficiently with differential IK and redundancy resolution for redundant systems. Moreover, the solution can warm-start the dynamics block. Fig. 7(*left*) shows a comparison of our approach with other existing methods. It is evident in the faster convergence of the ADMM (see Fig. 7) with three-block architecture solved in the consensus manner. Sequential variants tend to converge slower with the appropriate penalty parameters (ρ) used. On the other hand, the cost of SQP and Vanilla DDP methods converges comparable to the benchmark, but more iterations are needed for the constraint residuals to drop to a satisfactory threshold of 10^{-2}.

D. Constraint Satisfaction

Constraint residuals for each iteration in Fig. 7(right) are recorded, and the cost coefficients are tuned for each method. It is observed that residuals in the consensus ADMM with

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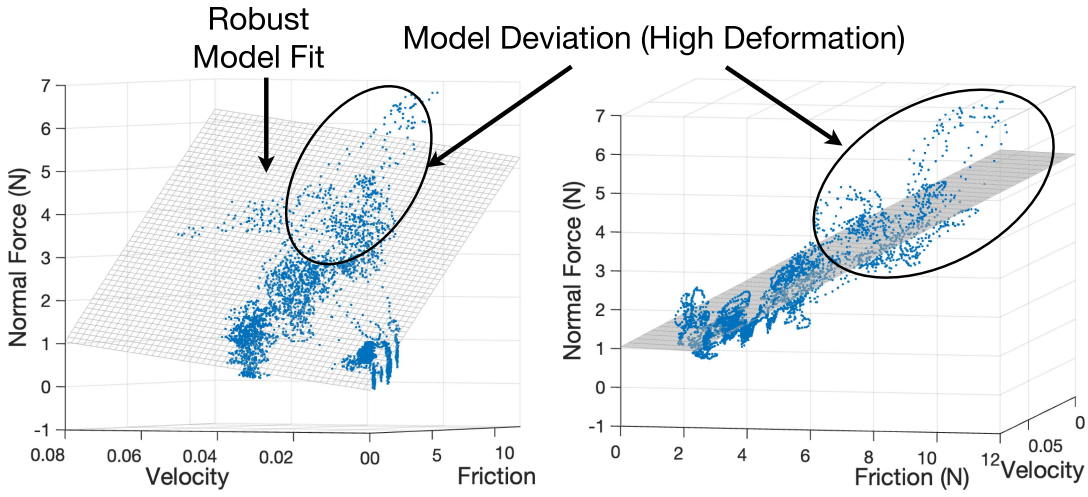


Fig. 8. Friction model validation and identification. $\mu = 0.4512$, $K_d = 13.1315$, and R-squared $= 0.9103$.

three-block architecture drop relatively faster than other methods. The SQP method initially started with a lower residual value and increased (and decreased again). This observation can be attributed to the preprocessing phase of the SQP solver to find an initial solution that is constraint satisfied regardless of the objective cost. The iLQR keeps the residual constant without significant improvements. It is due to the cost and constraint combined cost term, prioritizing cost over constraints.

E. Experimental Results

In our previous work [45], we demonstrated the validity of the contact model experimentally when the environmental platform is stationary. In this article, we extend it to a nonstationary environment where periodic disturbances are applied (1-Hz pulsations), as shown in Fig. 6. Moreover, we demonstrate the feasibility of the real-time deployment of the proposed framework. The experimental results can be summarized as follows:

1) open-loop trajectories in the presence and absence of environmental disturbances;
2) open-loop trajectories with low-level FC activated;
3) model-predictive controller with low-level FC in the presence and absence of disturbances.

F. System Identification of Material Properties

To experimentally validate the proposed soft contact model, parameters related to contact body material need to be identified, e.g., frictional coefficient and Young modulus. The friction coefficient is estimated by performing predetermined motions along the surface of the material surface while recording the force/torque data through an ATI mini45 sensor, which is attached to the end-effector in Fig. 6. The Young modulus is estimated through performing cyclic linear probing on the surface of the material with the same end-effector point geometry of a sphere (was tested on a material testing platform INSTRON©). It is performed through a nonlinear least squares estimator by using (1) and (6) and was used to estimate the frictional coefficients

TABLE I
ALGORITHMIC TIMES FOR THREE-BLOCK CONSENSUS, THREE-BLOCK SEQUENTIAL, TWO-BLOCK SEQUENTIAL, VANILLA DDP, AND SQP

Method	Time (s) $[10^3]$
Three-block CONSENSUS	3.0966
Three-block SEQUENTIAL	6.6121
Two-block SEQUENTIAL	16.620
VANILLA DDP	4.8436
SQP	6.0421

The whole trajectory was solved with constraints with equal number of time steps. Three-block consensus takes the shortest time, while two-block sequential takes the longest time.

by nonlinear least squares estimation as follows. Equation (20) summarizes the parameter estimation problem

$$\min_{E} \sum_{k=0}^{k=N} \|\bar{d}_k - \frac{9F_k^2}{16E^2R}^{\frac{1}{3}}\|_2 \quad \text{derived from (1)}$$

$$\min_{\mu,k_d} \sum_{k=0}^{k=N} \|\bar{d}_k - \mu F_z \left[1 + (2\nu - 1)\frac{3a^2}{10R^2}\right] \mathbf{n}_v \tag{20a}$$

$$- k_d \mathbf{v}_e\|_2 \quad \text{derived from (6).} \tag{20b}$$

Equation (20) presents the identification of parameters. Frictional force magnitude in the moving direction $\mathbf{F}_{\text{fric}}$, velocity magnitude $\mathbf{v}_e$, and normal contact force $\mathbf{F}_z$ are calculated from the collected data. A 3-D robust least squares approximation is fit with a logistic distance function in MATLAB©. This fitting is used to mitigate the sensitivity to the model deviation as the deformation increases (see Fig. 8). Identified Young modulus and friction coefficient were incorporated into the overall optimization in (11). Desired states to track are the desired end-effector position (x_e, y_e, z_e) and the desired normal contact force F_z.

The purpose of identifying material properties is two-fold: 1) to validate that the used models are well suited, and 2) to use in the TO framework to generate optimal open-loop

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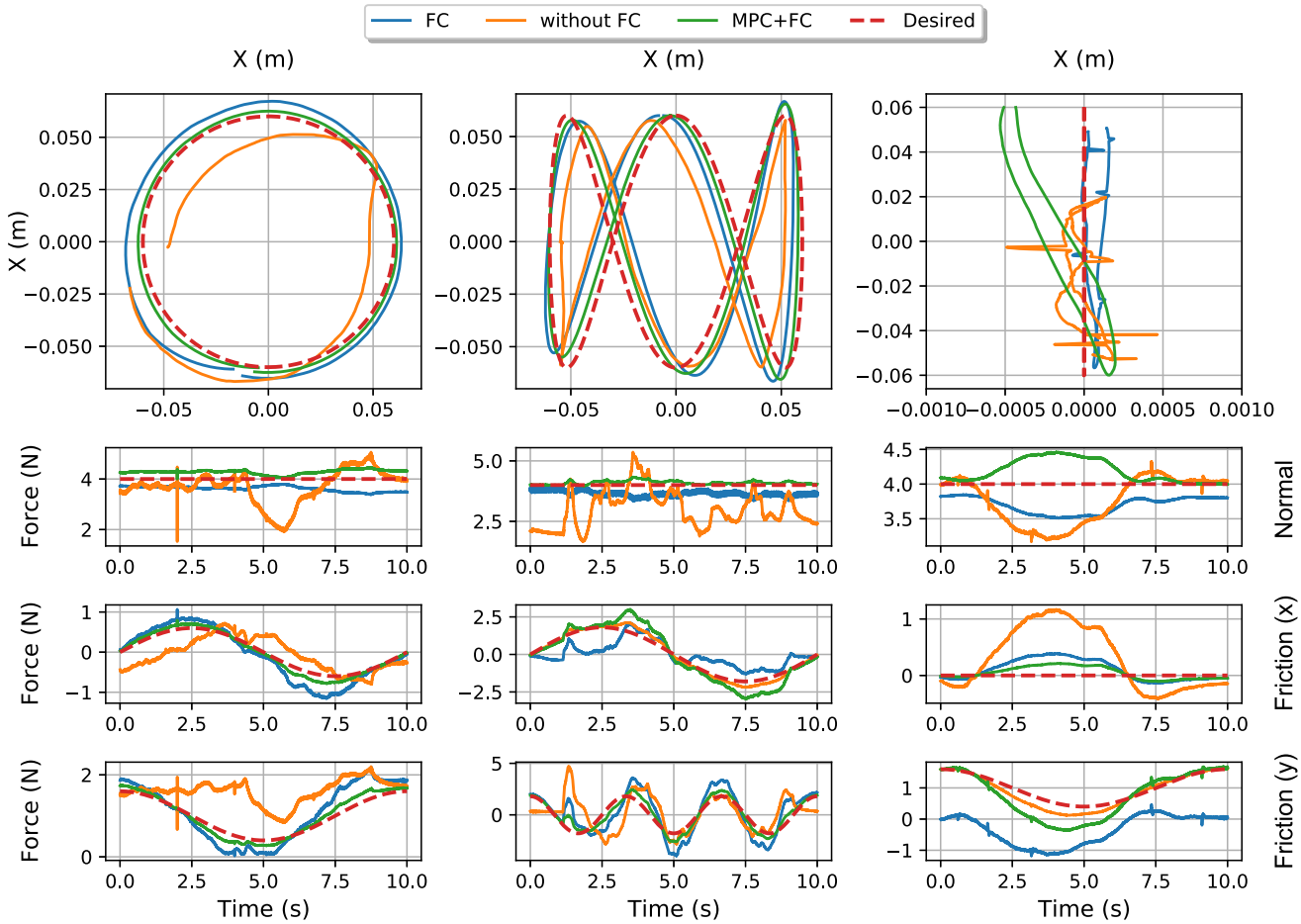


Fig. 9. *Top row:* Paths tracked by the end-effector versus commanded. *Bottom Row:* Comparison of ground truth of normal and frictional forces versus predicted by the model.

trajectories. Friction data fitting results are presented in Fig. 8. Data were fit with a resulting R-squared value of 0.9103. The frictional coefficient ($\mu = 0.4512$) and the damping coefficient ($k_d = 13.1315$) were identified.

It is observed that with the increase of normal force on the surface, the effects of deformation dominate the frictional forces. This phenomenon is due to the increased rolling friction and material-specific artifacts, e.g., nonuniformity in frictional coefficient and stress distribution. Moreover, the presence of fluids or any microparticular particles will increase the nonuniformity.

In the implementation, optimal state trajectories and inputs are generated through the optimization formulated in (11). Constraints were satisfied within 10^{-2} residual value violations in both primal and dual stopping criteria. Fig. 8 shows that the contact model used is valid for a range of normal forces. Therefore, the desired contact force is maintained within the valid bound of the friction model.

G. Open-Loop Trajectory Generation Without Environmental Disturbances

Open-loop trajectories generated from our TO method are used as reference trajectories for the experiments. Any mismatch in contact forces (e.g., those due to friction and deformation) would directly affect the motion and *vice versa*. It is observed that the control input solved via TO was able to track the motion ($\mathbf{x}_e^d$) and force profile ($\mathbf{F}_d$) significantly better than a position-controlled robot with force modulation, as shown in Fig. 9 . This is due to its open-loop compensation of frictional and centripetal forces encountered during the contact interaction. Table II presents the quantification of the results shown in Fig. 9. Although the tracking performance is not superior even with the proposed method, it is significantly improved with the existence of model inaccuracy caused by the soft material's unmodeled dynamics, stiction, and nonuniformity. We elaborate on the results in the following paragraphs.

As described in Section VII-G, the same open-loop full trajectories were executed with force control, as described in Section VI. Fig. 9 shows experimental results, which suggest that low-level force control alone (the blue trajectory) can improve the reference force tracking accuracy significantly. However, the reference motion tracking accuracy degrades due to the frictional forces encountered on the surface observed in the experimental results. The reference force and motion tracking performance improve with active force control and MPC (the green trajectory). Furthermore, Table II shows quantified results on multiple motion trajectories with different geometries with different curvatures. The motion accuracy is observed to be

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TABLE II
RSME FOR MOTION AND FORCE AND MOTION

	Motion Path 1		Motion Path 2		Motion Path 3	
	path (m)	force (N)	path (m)	force (N)	path (m)	force (N)
Pulsations, without FC	**0.0142**	2.723	**0.0264**	2.470	**0.0075**	1.377
Pulsations, with FC	0.0839	1.095	0.0757	1.289	0.0187	0.8450
Pulsations, MPC + FC	0.0258	**0.573**	0.0384	0.852	0.0153	**0.3230**

FC: admittance force control, MPC: model-predictive control.

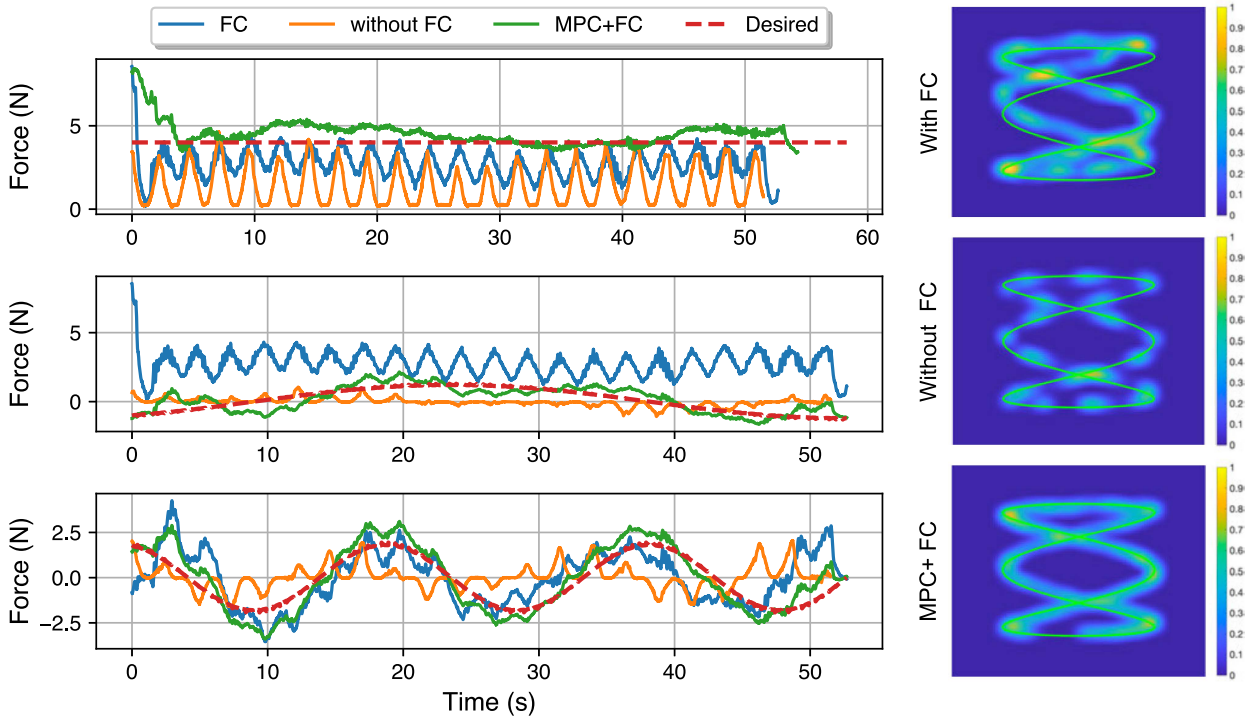


Fig. 10. *Left:* Normal and frictional forces with surface pulsations. Comparison of the activated FC (*blue*), inactivated force control (*orange*), and MPC and activated force control (*green*). *Right:* Normalized force and motion superimposed on a heat map. Note that motion pulsations of the experimental platform (see Fig. 6) cause the ripples observed in force profiles, for which the controller is attempting to compensate.

better without force control, but the force tracking is worse. On the other hand, with force control and MPC, both motion and force accuracy are improved as indicated by the corresponding RSME values. In Cartesian geometries that contain sharp curves (e.g., rectangular geometry), the centripetal force component is an addition to the frictional force in contact force compensation; the velocity at the corners needs to vary to maintain the path and the contact, as illustrated in Fig. 3. Failing to compensate for it could result in sliding and deviating from the desired motion (the *without FC* case in Fig. 9). Only friction force needs to be compensated in a straight line, while both centripetal and frictional components are present in a curved geometry.

H. MPC With Low-Level FC Under Periodic Environmental Disturbances

In Section VII-G, it was shown that force control and MPC could improve both force and motion tracking accuracy. However, the environment is subject to motion disturbances in a realistic setting. To cope with such disturbances and compensate for frictional forces simultaneously, we experimentally show the efficacy of the proposed TO method. Fig. 10 illustrates the force and motion tracking accuracy when under force control, without force control, and with force control, and MPC. Similar to the static case, low-level force control improves force tracking, but the "reactive" nature can be observed in the force tracking (in Fig. 10, right column and second row subplot). As a result, the MPC with contact model information improves both force and motion tracking accuracy significantly. Moreover, the system remains stable with externally induced motion disturbances and model uncertainties. Such a success can be attributed to the lower level admittance controller and the model-based TO run in a MPC fashion. Experiments were run on multiple Cartesian trajectories to validate on varying geometries, and quantified results are reported in Table III.

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TABLE III
RSME FOR MOTION AND FORCE AND MOTION

	Motion Path 1		Motion Path 2		Motion Path 3	
	path (m)	force (N)	path (m)	force (N)	path (m)	force (N)
without FC	0.284	1.042	0.392	1.328	0.302	0.891
with FC	0.0183	0.384	0.0258	0.481	0.0172	0.0273
MPC + FC	**0.0164**	**0.283**	**0.0191**	**0.319**	**0.0233**	**0.0084**

FC: admittance force control, MPC: model-predictive control; Motion path 1: Circle, Motion path 2: Eight, and Motion path 3: line.

I. Discussion and Limitations

The experiments and results demonstrated the importance of incorporating contact in the TO framework for better force and motion tracking accuracy. In safety-critical applications, stiff position-controlled robots are not desired as they could raise safety concerns. Contact-model-based force-controlled control architectures could provide safe and improved performance. Our static environment results exhibit better performance compared with a motion-induced dynamic environment. Fitting the environment motion disturbance (e.g., breathing and heartbeat) to a parametric model can further improve the tracking performance. In our current work, the robot's impedance is set to a predefined mode that can be limiting. For example, a high-impedance robot is more suited for tasks requiring more motion accuracy over force and *vice versa*. Adapting the robot's impedance depending on tasks and applications can further improve safety, force, and motion tracking accuracy.

While our work presents a method for force and motion TO, we acknowledge multiple limitations pertaining to the application and computational techniques. For example, the contact model we use is only valid locally. For large deformable bodies, the contact model will require a more computationally expensive method (e.g., finite-element method). Furthermore, the range of force magnitudes was maintained throughout the experiments to be within a local range (in the linear range of Fig. 8). To compensate for the friction in high deformation and lubrication cases, additional factors of the deformation friction will have to be learned or modeled.

Although this work achieved MPC online planning, there are more promising TO parallelization mechanisms [91], [92], [93] that can further improve the computational performance.

1) In the DDP backward pass, a computation speedup is achievable through a parallelization method of Riccati recursion [81].
2) In the DDP forward pass, computation speedup can be targeted by first proposing an approximate physics model, where a computationally cheap estimate of a coarse model can be evaluated [94]. Then, the generated coarse trajectory can be used as an initial seed of the TO with a fine-grained model. This coarse-fine problem can be solved in an iterative procedure.
3) Finally, these mechanisms above will suit a paralleled ADMM implementation on GPU processors.

Although this article does not focus on TO parallelization mechanisms, these potential directions are worth to be reported and can be insightful to the deformable tissue manipulation and medical robotics community.

VIII. CONCLUSION

In automation tasks requiring physically soft tissue contact, it is paramount to design soft contact interaction models where controllers can be designed to guarantee safety performance. Contact modeling is crucial in correctly identifying the contact material and performing mundane tasks such as incisions along given paths and motion disturbance compensation. This article presented a coherent framework for simultaneous motion and force modulation on compliant surfaces. Moreover, we presented a distributed (ADMM) real-time framework executed in an MPC fashion capable of handling state, control, and contact constraints. Furthermore, we incorporated a soft contact dynamical model into the TO. Results proved that motion and force tracking accuracy is significantly improved in both the static and dynamic environments. Potential applications of this work include contact manipulation in soft tissues or safety-critical environments.

Trajectories solved from the TO were experimentally validated on a soft surface (EcoFlex©) with the aid of a robot manipulator with an attached spherical shaped tool-tip. Surface material properties were estimated and further used in generating optimal trajectories. Experiments were performed on a static and a motion-induced dynamic environment. Results of MPC, with and without force control, were presented. Ground-truth forces were obtained using a force–torque sensor (ATI mini45) and compared against the obtained results. MPC with force control was able to track both motion and force both in a static and a dynamic environment with significant improvements. Results and discussion conclude that model-based contact modeling and hierarchical TO (e.g., low level and high level) provide a better alternative for safe simultaneous force and motion generation.

The future extension of this work is to improve the generability (i.e., "richness") of the contact model to adapt to a wide range of material properties. Furthermore, the real-time estimation of the contact model properties can improve the adaptability of the planning framework. Moreover, we intend to extend the work to plan trajectories in 3-D surfaces to demonstrate practical applications such as planning robotic incisions on a human body.

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APPENDIX A
DEFORMABLE PATCH CONTACT MODEL

In this appendix, we provide the details on several stress distributions.

1) Normal stress distribution σ_z:

$$\frac{\sigma_z}{p_m} = -\frac{3}{2}\left(1-\frac{r^2}{a^2}\right)^{\frac{1}{2}} \quad (r \le a). \tag{21}$$

2) Radical stress distribution σ_r:

$$\frac{\sigma_r}{p_m} = \frac{2\nu-1}{2}\frac{a^2}{r^2}\left[1-\left(1-\frac{r^2}{a^2}\right)\right] - 3\nu\left(1-\frac{r^2}{a^2}\right)^{\frac{1}{2}} \quad (r \le a).$$

3) Hoop stress distribution σ_θ:

$$\frac{\sigma_\theta}{p_m} = \frac{1-2\nu}{2}\frac{a^2}{r^2}\left[1-\left(1-\frac{r^2}{a^2}\right)\right] - \frac{3}{2}\left(1-\frac{r^2}{a^2}\right)^{\frac{1}{2}} \quad (r \le a)$$

where $p_m = F/(\pi a^2)$ is the average stress applied in contact part by manipulation and $a = \sqrt{Rd}$ is the radius of contact area (refer to Fig. 2).

The transformation matrix T is

$$T = \begin{bmatrix} c\theta & s\theta & 0 \\ -s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

APPENDIX B
THREE-BLOCK SEQUENTIAL ADMM

Instead of establishing a consistency constraint between the decision variables from the IK subblock and the projection subblock (i.e., $\hat{\mathbf{q}} = \bar{\mathbf{q}}$ in Formulation 2), the sequential ADMM enforces an equality between DDP subblock and IK subblock, i.e., $\mathbf{q} = \hat{\mathbf{q}}$, as shown in Formulation 3.

Different from the consensus variant, the original optimization problem is separated into the following.

1) Whole-body dynamics subblock:

$$\mathcal{L}_{\text{wbd}}(\boldsymbol{\phi}, \hat{\boldsymbol{\phi}}, \bar{\boldsymbol{\phi}}, \mathbf{v}) = \sum_{i=0}^{N}(\delta\mathbf{F}[i]^T\mathbf{Q}_F\,\delta\mathbf{F}[i] + \mathbf{u}[i]^T\mathbf{R}\mathbf{u}[i]) + \frac{\rho_j}{2}\|\mathbf{q}-\hat{\mathbf{q}}+\mathbf{v}_{\text{ik}}^k\|_2^2 + \frac{\rho_j}{2}\|\mathbf{q}-\bar{\mathbf{q}}+\mathbf{v}_j^k\|_2^2 + \frac{\rho_u}{2}\|\mathbf{u}-\bar{\mathbf{u}}+\mathbf{v}_u\|_2^2 + \frac{\rho_f}{2}\|\boldsymbol{\lambda}-\bar{\boldsymbol{\lambda}}+\mathbf{v}_f\|_2^2.$$

2) Inverse kinematics subblock:

$$\mathcal{L}_{\text{ik}}(\boldsymbol{\phi}, \hat{\boldsymbol{\phi}}, \mathbf{v}) = \sum_{i=0}^{N} W_p\left\|\mathcal{F}\mathcal{K}(\hat{\mathbf{q}}[i]) - \mathbf{x}_e^d[i]\right\|_2 + \frac{\rho_j}{2}\|\mathbf{q}-\hat{\mathbf{q}}+\mathbf{v}_{\text{ik}}\|_2^2.$$

Formulation 3 Distributed Constrained Optimization (Sequential)

$$\begin{aligned}
&\text{(tracking task)} && \min_{\boldsymbol{\phi},\hat{\boldsymbol{\phi}},\bar{\boldsymbol{\phi}}} \sum_{i=0}^{N} \delta\mathbf{F}[i]^T\mathbf{Q}_F\,\delta\mathbf{F}[i] + \mathbf{u}[i]^T\mathbf{R}\mathbf{u}[i] \\
&&& \quad + W_p\left\|\mathcal{F}\mathcal{K}(\mathbf{q}[i]) - \mathbf{x}_e^d[i]\right\|_2^2 \\
&&& \quad + I_{\mathcal{J},\mathcal{U},\mathcal{F}}(\bar{\mathbf{q}}[i], \bar{\mathbf{u}}[i], \bar{\boldsymbol{\lambda}}[i]) \\
&\text{(Variables-DDP)} && \boldsymbol{\phi}[i] = [[\overbrace{\mathbf{q}[i]\ \dot{\mathbf{q}}[i]\ \mathbf{F}_e[i]}^{\mathbf{q}[i]}]^T, \mathbf{u}[i]^T]^T \\
&\text{(Variables-IK)} && \hat{\boldsymbol{\phi}}[i] = \hat{\mathbf{q}}[i]^T \\
&\text{(Variables-Proj)} && \bar{\boldsymbol{\phi}}[i] = [\bar{\mathbf{q}}[i]^T, \bar{\mathbf{u}}[i]^T, \bar{\boldsymbol{\lambda}}[i]^T]^T \\
&&& \forall i = 0, 1, \ldots, N-1 && \text{(22a)} \\
&\text{(Dynamics)} && \text{s.t.}\ \ \mathbf{x}[i+1] = \mathcal{F}(\mathbf{x}[i], \mathbf{u}[i]) && \text{(22b)} \\
&\text{(Initial condition)} && \mathbf{x}[0] = \mathbf{x}_0 && \text{(22c)} \\
&\text{(Consistency constraints)} && \begin{cases} \mathbf{q} = \hat{\mathbf{q}} \\ \mathbf{q} = \bar{\mathbf{q}} \\ \mathbf{u} = \bar{\mathbf{u}} \\ \boldsymbol{\lambda} = \bar{\boldsymbol{\lambda}} \end{cases} && \text{(22d)}
\end{aligned}$$

3) Projection subblock:

$$\mathcal{L}_{\text{proj}}(\boldsymbol{\phi}, \bar{\boldsymbol{\phi}}, \mathbf{v}) = \sum_{i=0}^{N} I_{\mathcal{J},\mathcal{U},\mathcal{F}}(\bar{\mathbf{q}}[i], \bar{\mathbf{u}}[i], \bar{\boldsymbol{\lambda}}[i]) + \frac{\rho_j}{2}\|\mathbf{q}-\bar{\mathbf{q}}+\mathbf{v}_j\|_2^2 + \frac{\rho_u}{2}\|\mathbf{u}-\bar{\mathbf{u}}+\mathbf{v}_u\|_2^2 + \frac{\rho_f}{2}\|\boldsymbol{\lambda}-\bar{\boldsymbol{\lambda}}+\mathbf{v}_f\|_2^2.$$

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